Type System v1.5 Inference

Hindley-Milner Meets Gradual Typing in Lateralus

Lateralus Language

bad-antics · April 2026 · Lateralus Language Research

1. Type System v1.5: Feature Overview

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

Performance of the type checker matters for developer experience. In benchmark studies of real Lateralus codebases, type checking accounts for roughly 30% of total compilation time. The remaining time is split between parsing (15%), name resolution (10%), code generation (35%), and linking (10%). The type checker has been optimized with a persistent hash-consing scheme for type expressions, which eliminates redundant allocations for commonly-occurring types like 'int' and 'str'. With hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

Implementation Details

The integration between the type system and the module system is a critical architectural concern. Modules export typed interfaces, and importing code sees only the declared interface -- not the implementation. This means that if a module's implementation changes but its interface stays the same, dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

Testing the type system is itself a substantial engineering challenge. The test suite includes thousands of positive tests (programs that should type-check) and negative tests (programs that should not). Negative tests are particularly important: each one verifies not just that the type checker rejects the program, but also that it produces the correct error message at the correct source location. The test framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Type inference walkthrough
```

```
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

2. Type Expressions and Syntax

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

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Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

3. Algorithm W: Core Inference

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}
```

```
let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

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Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

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4. Unification Algorithm

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framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

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Annotation optional optional none

Blame static labeled none

5. Let Polymorphism and Generalization

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

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6. The Occurs Check

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
```

```
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
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Error location compile compile+run runtime

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7. Bidirectional Type Checking

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    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
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```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

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Error location compile compile+run runtime

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8. Type Classes: Traits

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// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
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let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
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9. Trait Implementations

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

10. Numeric Type Hierarchy

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

11. Generic Functions

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
```

```
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str

println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

12. Higher-Kinded Types (Planned)

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

13. Effect System Integration

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

14. Row Polymorphism

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

15. Gradual Typing Integration

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

16. Type Inference for Pipelines

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
```

```
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

17. Error Message Quality

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

18. Async Type Inference

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Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes

higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

19. Principal Types

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

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framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

20. Type Aliases and Newtypes

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),

    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

21. Subtyping and Variance

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
```

```
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

22. Type-Level Computation

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

23. Inference Debugging: Type Holes

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

24. Type Coverage Metrics

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

25. Design Choices Comparison

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

26. Inference Limitations

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated

by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

Performance of the type checker matters for developer experience. In benchmark studies of real Lateralus codebases, type checking accounts for roughly 30% of total compilation time. The remaining time is split between parsing (15%), name resolution (10%), code generation (35%), and linking (10%). The type checker has been optimized with a persistent hash-consing scheme for type expressions, which eliminates redundant allocations for commonly-occurring types like 'int' and 'str'. With hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

27. Soundness of the Type System

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

28. Completeness of Inference

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]

let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

29. Future: Dependent Types

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Trait-based polymorphism

trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }
```

```
fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

30. Type System Summary

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

31. Type System: Interaction with Macros

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

32. Type System: Interface Stability

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

33. Type System: Advanced Constraints

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

34. Type System: Totality and Termination

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice

to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

Performance of the type checker matters for developer experience. In benchmark studies of real Lateralus codebases, type checking accounts for roughly 30% of total compilation time. The remaining time is split between parsing (15%), name resolution (10%), code generation (35%), and linking (10%). The type checker has been optimized with a persistent hash-consing scheme for type expressions, which eliminates redundant allocations for commonly-occurring types like 'int' and 'str'. With hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

Implementation Details

The integration between the type system and the module system is a critical architectural concern. Modules export typed interfaces, and importing code sees only the declared interface -- not the implementation. This means that if a module's implementation changes but its interface stays the same, dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

Testing the type system is itself a substantial engineering challenge. The test suite includes thousands of positive tests (programs that should type-check) and negative tests (programs that should not). Negative tests are particularly important: each one verifies not just that the type checker rejects the program, but also that it produces the correct error message at the correct source location. The test framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
```

```
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

35. Type System: Generic Specialization

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated

by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

Implementation Details

The integration between the type system and the module system is a critical architectural concern. Modules export typed interfaces, and importing code sees only the declared interface -- not the implementation. This means that if a module's implementation changes but its interface stays the same, dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

Testing the type system is itself a substantial engineering challenge. The test suite includes thousands of positive tests (programs that should type-check) and negative tests (programs that should not). Negative tests are particularly important: each one verifies not just that the type checker rejects the program, but also that it produces the correct error message at the correct source location. The test framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

36. Type System: Effect Polymorphism

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

Performance of the type checker matters for developer experience. In benchmark studies of real Lateralus codebases, type checking accounts for roughly 30% of total compilation time. The remaining time is split between parsing (15%), name resolution (10%), code generation (35%), and linking (10%). The type checker has been optimized with a persistent hash-consing scheme for type expressions, which eliminates redundant allocations for commonly-occurring types like 'int' and 'str'. With hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

Implementation Details

The integration between the type system and the module system is a critical architectural concern. Modules export typed interfaces, and importing code sees only the declared interface -- not the implementation. This means that if a module's implementation changes but its interface stays the same, dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

37. Type System: Variance in Practice

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

The integration between the type system and the module system is a critical architectural concern. Modules export typed interfaces, and importing code sees only the declared interface -- not the implementation. This means that if a module's implementation changes but its interface stays the same, dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

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program, but also that it produces the correct error message at the correct source location. The test framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

38. Type System: Higher-Kinded Encoding

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}
```

```
let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

39. Type System: Proof Obligations

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }
```

```
fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

40. Type System: Incremental Checking

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

41. Type System: Parallel Checking

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes

higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

The integration between the type system and the module system is a critical architectural concern. Modules export typed interfaces, and importing code sees only the declared interface -- not the implementation. This means that if a module's implementation changes but its interface stays the same, dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

42. Type System: Code Generation

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the

expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

Performance of the type checker matters for developer experience. In benchmark studies of real Lateralus codebases, type checking accounts for roughly 30% of total compilation time. The remaining time is split between parsing (15%), name resolution (10%), code generation (35%), and linking (10%). The type checker has been optimized with a persistent hash-consing scheme for type expressions, which eliminates redundant allocations for commonly-occurring types like 'int' and 'str'. With hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

Implementation Details

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Testing the type system is itself a substantial engineering challenge. The test suite includes thousands of positive tests (programs that should type-check) and negative tests (programs that should not). Negative tests are particularly important: each one verifies not just that the type checker rejects the program, but also that it produces the correct error message at the correct source location. The test

framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

43. Type System: Optimization Passes

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

44. Type System: Documentation Generation

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every

well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}
```

```
// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

45. Type System: IDE Protocol

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)

println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

46. Type System: Error Recovery

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str

let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

47. Type System: Cross-Module Checking

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}
```

```
// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

48. Type System: Appendix -- Rule Reference

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

49. Type System: Interaction with Macros

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

50. Type System: Interface Stability Guarantees

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

51. Type System: Advanced Constraint Solving

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit
```

```
// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

52. Type System: Totality and Termination

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
```

```
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}

// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

53. Type System: Generic Specialization

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

54. Type System: Effect Polymorphism Deep Dive

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

55. Type System: Variance in Practice

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framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Recursive type inference

fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

56. Type System: Higher-Kinded Type Encoding

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

57. Type System: Proof Obligations

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
```

```
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}

// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

58. Type System: Incremental Type Checking

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Performance of the type checker matters for developer experience. In benchmark studies of real Lateralus codebases,

type checking accounts for roughly 30% of total compilation time. The remaining time is split between parsing (15%), name resolution (10%), code generation (35%), and linking (10%). The type checker has been optimized with a persistent hash-consing scheme for type expressions, which eliminates

redundant allocations for commonly-occurring types like 'int' and 'str'. With hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

Implementation Details

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

59. Type System: Parallel Type Checking

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the

expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

60. Type System: Type-Directed Code Generation

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice

to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

61. Type System: Optimizations Enabled by Types

The Lateralus language type system represents a careful synthesis of ideas from programming language research

spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit
```

```
// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

62. Type System: Documentation Generation

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
```

```
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}

// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

63. Type System: IDE Protocol Integration

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]

let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

64. Type System: Advanced Error Recovery

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is

acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

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Implementation Details

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

65. Type System: Cross-Module Type Checking

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

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Implementation Details

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

66. Type System: Plugin Architecture

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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Implementation Details

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit
```

```
// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

67. Type System: Testing the Type Checker

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
```

```
    let result: list[(A,B)] = []
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}

// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

68. Type System: Benchmark Suite

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}

let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

69. Type System: Regression Testing

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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Implementation Details

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

70. Type System: Fuzzing the Type Checker

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```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

71. Type System: Formal Specification

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit

// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

72. Type System: Reference Implementation

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```
// Generic function inference
fn zip[A, B](as: list[A], bs: list[B]) -> list[(A,B)] {
    let result: list[(A,B)] = []
```

```
    for i in range(min(len(as), len(bs))) {
        result.push((as[i], bs[i]))
    }
    return result
}

// Inferred at call site:
let z = zip([1,2,3], ["a","b","c"])
// A=int (from [1,2,3]), B=str (from ["a","b","c"])
// z: list[(int, str)]
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

73. Type System: Evolution and Versioning

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Pipeline type propagation
fn parse_line(s: str) -> list[int] {
    return s.split(",") |> map(fn(x) { int(x.strip()) })
}
```

```
let lines = read_file("data.csv").split("\n")
let all_ints = lines |> map(parse_line)  // list[list[int]]
let flat     = all_ints |> flatten()     // list[int]
let sorted   = flat |> sort()            // list[int]
let unique   = sorted |> dedup()         // list[int]
println(unique.len(), "unique values")
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

74. Type System: Community Feedback Loop

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

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```
// Trait-based polymorphism
trait Printable { fn print_repr(self) -> str }
trait Comparable { fn compare(self, other: Self) -> int }

fn top_n[T: Printable + Comparable](items: list[T], n: int) -> list[T] {
    let sorted = items |> sort_by(fn(a, b) { a.compare(b) })
    return sorted |> take(n)
}

// Works for any T that implements both traits:
top_n([5,3,8,1,9,2,7], 3)   // [1,2,3]: int satisfies both
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

75. Type System: Adoption Patterns

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

Performance of the type checker matters for developer experience. In benchmark studies of real Lateralus codebases, type checking accounts for roughly 30% of total compilation time. The remaining time is split between parsing (15%), name resolution (10%), code generation (35%), and linking (10%). The type checker has been optimized with a persistent hash-consing scheme for type expressions, which eliminates redundant allocations for commonly-occurring types like 'int' and 'str'. With hash-consing, type expressions that appear multiple times in the program share a single representation in memory, reducing both allocation pressure and comparison cost.

Implementation Details

The integration between the type system and the module system is a critical architectural concern. Modules export typed interfaces, and importing code sees only the declared interface -- not the implementation. This means that if a module's implementation changes but its interface stays the same, dependent code does not need to be re-type-checked. The compiler exploits this property for incremental compilation: when a file changes, only that file and its transitive dependents that actually use the changed interface are re-checked. In large codebases, this can reduce re-check time by 90% compared to full re-checking.

Testing the type system is itself a substantial engineering challenge. The test suite includes thousands of positive tests (programs that should type-check) and negative tests (programs that should not). Negative tests are particularly important: each one verifies not just that the type checker rejects the program, but also that it produces the correct error message at the correct source location. The test framework compares the actual error output against a golden file, which makes it easy to detect regressions where a bug fix in one area accidentally changes error messages in another.

```
// Recursive type inference
fn sum_tree(node: Tree[int]) -> int {
    match node {
        Tree.Leaf(v)        => v,
        Tree.Branch(l, r)   => sum_tree(l) + sum_tree(r),
    }
}
// Type: fn(Tree[int]) -> int (fully inferred from match)

let tree = Tree.Branch(
    Tree.Branch(Tree.Leaf(1), Tree.Leaf(2)),
    Tree.Leaf(3)
)
println(sum_tree(tree))  // 6
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

76. Type System: Summary and Future Work

The Lateralus language type system represents a careful synthesis of ideas from programming language research spanning four decades. The core inference engine uses Hindley-Milner Algorithm W, which guarantees that every well-typed program receives its most general -- or principal -- type. This ensures that functions written without any type annotations are still precisely typed, and that the compiler can catch type errors even in code that was never annotated by the programmer. The inference algorithm runs in near-linear time for typical programs, making interactive type checking fast enough for use in IDE integrations where sub-100ms response times are required.

Practical considerations drove every design decision in the type system. For instance, the choice to use structural typing for function types -- where any function with the right parameter and return types is acceptable regardless of how it was defined -- was made because it reduces boilerplate and makes higher-order programming more ergonomic. The choice to use nominal typing for struct and enum definitions was made because it prevents accidental substitution of structurally-similar but semantically-different types, which is a common source of subtle bugs. These two choices work together to give programmers flexibility where it helps and safety where it matters.

The compiler's error messages are a first-class feature of the type system implementation. When a type mismatch occurs, the compiler must report not just what went wrong but where -- including the source locations of both the expected and actual types -- and ideally suggest how to fix it. To support this, the

type checker maintains provenance information throughout inference: each type variable carries the source location where it was introduced, each unification constraint records both sides and their sources, and the substitution is applied lazily so that error messages can show the original type expressions rather than fully-substituted forms.

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```
// Type inference walkthrough
let pair = (42, "hello")   // pair: (int, str)
let (n, s) = pair          // n: int, s: str
let doubled = n * 2        // doubled: int
let upper = s.to_upper()   // upper: str
println(doubled, upper)    // type: (int, str) -> unit
```

```
// All types fully inferred; no annotations written
// Algorithm W assigns fresh vars, unifies, substitutes
```

Feature Static Gradual Dynamic

Null safety compile-time cast+runtime runtime

Type inference full (HM) partial (HM+any) none

Perf overhead 0% 2-8% 15-40%

Error location compile compile+run runtime

Annotation optional optional none

Blame static labeled none

Lateralus is an open-source, zero-dependency programming language. Project home: https://lateralus.dev. Source: github.com/bad-antics/lateralus-lang. Released under CC BY 4.0.